# LAKSHIMI RAMAN S

Database Administrator / Backend Engineer / Distributed Systems Engineer

☒ lakshimiramans@gmail.com **in** lakshimi-raman-s **GH** LakshimiRam-073 • Chennai, India

## SUMMARY

Database Administrator on Zoho's SaaS Framework team (15 engineers), working on PostgreSQL and MySQL infrastructure, automated failover, zero-downtime upgrades, replication tooling, and on-call production support for 50+ products and 50M+ tenants across 900+ clusters and 54 data centers. Strong in Go, Java, C, AWS, system design, and database internals.

## EXPERIENCE

**Database Administrator, SaaS Framework Team**
**Zoho Corporation** Jun 2025 – Present
*Chennai, India*

- Administer 900+ PostgreSQL and MySQL clusters across 54 data centers for 50+ Zoho applications, with multi-tenant provisioning for 50M+ tenants.
- Cut cluster failover from 1–2 minutes to roughly 5 seconds with a high-availability failover system in Go using witness-quorum consensus, removing manual split-brain checks during network partitions.
- Architected zero-downtime blue-green PostgreSQL upgrades, cutting spare capacity from 6 nodes per service to 2 and saving roughly 40 high-spec nodes per major upgrade.
- Engineered a distributed replication pipeline on PostgreSQL logical replication with a custom WAL decoder for cross-service DDL replay, eliminating schema-drift bugs the prior ORM-based decoder overlooked and running roughly 2.5x faster.
- Developed a cloud-scale backup and disaster recovery framework handling roughly 2 TB/week per cluster across PostgreSQL, MySQL, ClickHouse, etcd, Tarantool, and Prometheus.
- Optimized backend query routing by productionizing ProxySQL with TLS for SaaS clusters, reducing master load by roughly 40 percent on read-heavy applications.
- Resolved on-call production incidents across 900+ clusters in database (replication lag, deadlocks, query regressions, MVCC bloat) and network (TLS, VLAN, ProxySQL routing) layers via Prometheus, Grafana, and engine internals.
- Delivered zero-downtime VLAN migration for live production clusters by promoting a secondary as cutover target without breaking app connections.
- Leading a graph database POC evaluating DGraph, JanusGraph (Cassandra), and SurrealDB to give 50+ app teams native graph support inside our cluster topology.

**Additional: PGConf India 2026, Bangalore** Attendee
Networked with the ClickHouse co-founder; brought back PostgreSQL scaling patterns from 5+ companies. Reads database papers (Spanner, GFS, Neon).

## PROJECTS

**Cluster Management System** Go, etcd
Built a backend orchestration service for MySQL and PostgreSQL clusters with witness-based quorum, automated failover, VIP migration, rolling upgrades, resumable workflows, and ProxySQL or PgBouncer integration; manages 900+ production clusters.

**Internal-workings** Go, C
Repo of systems primitives built from scratch: a paged **disk manager** in Go (bitmap free-space, file/mmap storage, page sync); a **linear-probing hash map** in Go; a 200-line C **memory allocator** (malloc/free with sbrk, mmap, free-lists); a C **network socket** server. github.com/LakshimiRam-073/Internal-workings

**Personal Tech Blog** stupidnotes.in
6+ long-form articles on database internals, system design, and low-level engineering.

## KEY ACHIEVEMENTS

**Google Foobar – Level 4**
Reached Level 4 of Google's invite-only algorithmic challenge.

**Smart India Hackathon**
Finalist in the national-level innovation hackathon.

**BIT Virupaksha**
Finalist in the ML Hackathon at Bannari Amman Institute of Technology.

## EDUCATION

**B.E. Electronics and Communication Engineering**
Sri Krishna College of Engineering and Technology Oct 2021 – May 2025
Coimbatore • CGPA 8.44

## SKILLS

**Languages** — Go · Java · C · C++ · Python · SQL · Bash · Data Structures & Algorithms

**Databases** — PostgreSQL · MySQL · ClickHouse · Redis · etcd · Tarantool · DGraph · Query Optimization · Indexing · MVCC · WAL · Storage Engines (B+ Tree, LSM)

**Distributed Systems & Backend** — REST APIs · gRPC · Microservices · System Design · Consensus (Raft) · High Availability · Fault Tolerance · Concurrency · Data Consistency · Sharding · Logical Replication · Blue-Green Deployment

**Cloud & Tools** — AWS · Linux · Docker · Firecracker · ProxySQL · PgBouncer · Caching · Prometheus · Grafana · Observability · CI/CD · TLS/mTLS

## INTERESTS

**Database Internals**
Reads research papers (Spanner, GFS, Neon) and writes long-form articles on storage and replication.

**System Design**
Studies large-scale architectures (sharding, consensus, replication trade-offs) and prototypes them in side projects.

**Low-level Engineering**
Building allocators, decoders, and tooling in C and Go from first principles.

## LANGUAGES

**English** Professional • **Tamil** Native